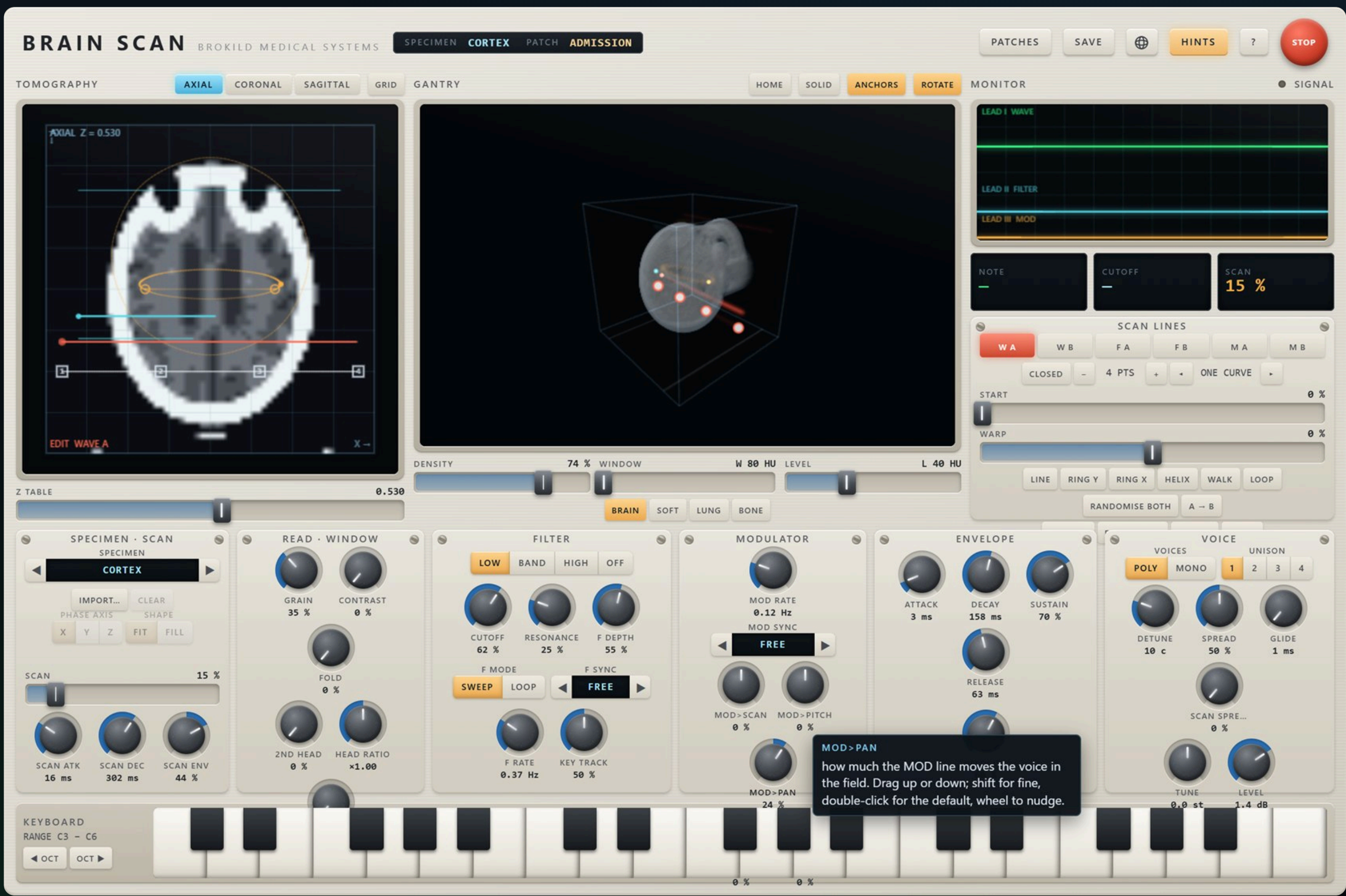
BROKILD · WINDOWS VST3 · OWNER'S MANUAL

BRAIN SCAN

A synth that stores no waveforms. A specimen is a volume; a scan line is a curve through it; one cycle of the sound is the field along that curve. The filter's cutoff and the modulator are the same thing read slower. And SCAN blends the geometry of two lines, so halfway between two sounds is tissue neither of them has visited.



BRAIN SCAN

fifteen specimens · three scanners · six lines · tomography · gantry · world rack

BROKILD

free · VST3 + standalone · 64-bit

01 The answer and the question

A wavetable synth stores a waveform and reads it with a phasor. To move between two sounds it crossfades two answers. Brain Scan keeps the idea of a stored thing and changes what is stored, and the whole instrument follows from that one change.

A specimen is a volume. Not a table of samples but a scalar field `V(x,y,z)` filling a cube — a density, exactly like the numbers a scanner records. Fifteen of them ship, in two families. Nine are *phantoms* — calibration objects, each a formula and each a promise: six give ordinary, useful waveforms, three are made of edges, spikes and beats. Six are *bodies*, built in Hounsfield units the way a scanner would record them: a head, the same head as bare skull, a chest, three lumbar vertebrae, a thigh, and a jaw with its teeth.

A scan line is a curve through it. Two to sixteen control points with a smooth curve through them, open or closed. One cycle of the waveform is the field read along that curve: `w(s) = V(g(s))`, where `s` is the phase. Drive the curve straight across the cube and you read the specimen's natural waveform. Bend it, loop it, tangle it, and you read something the specimen contains but nobody put there on purpose.

What the geometry gives you for nothing

- A **closed loop** has no edge, so the cycle joins itself: soft.
- An **open segment** jumps between its ends once per cycle: bright, saw-like. That jump is band-limited so it does not alias.
- A **corner** is a kink at a fixed phase.
- A **self-crossing** is one value visited at two phases.
- **WARP** reads the same shape unevenly: phase distortion, free.
- Move the whole line and the timbre evolves while the instrument stays itself.

The claim, and how it is different

SCAN blends the positions, not the samples. Every scanner has two anchor lines, A and B. At SCAN 50 % the instrument does not mix the two waveforms; it builds the line halfway between the two *curves* and reads the field once, along that. A wavetable crossfades two answers. This moves the question.

> **It is measurable, and it is measured.** Take two pulses of duty 0.23 and 0.77. Both have a null at the fifth harmonic, so any crossfade of them keeps that null: the bench reads it at −66 dB. The midway *path* reads a pulse of duty 0.50, whose fifth harmonic is −14.0 dB, exactly the formula. Fifty decibels of difference, from the same two endpoints.

The filter and the modulator are the same mechanism read slower. The cutoff is a scan line traversed once per note, or looping; so is the modulator. Three scanners, six lines, one law, and one SCAN control that moves all three at once.

02 Thirty seconds of instructions

Play something. The keyboard along the bottom is playable with the mouse, and the computer keys work too: `z x c v` for the lower octave, `q w e r` for the one above. The instrument opens on a patch called ADMISSION.

Turn SCAN. It is the wide slider under SPECIMEN, on the left of the desk. Watch the bright line on the slice move between its two dim anchors while the sound travels. That is the whole instrument in one gesture.

Change the specimen. The window at the top left of the desk steps through the fifteen. The gantry and the slice reload; the lines stay where they are, so the same drawing reads a different body.

Draw. Pick **W A** in the SCAN LINES panel, then work on the tomography slice: drag a numbered point to move it, click bare tissue to add one, alt-click a point to take it away. A point keeps its depth when dragged; the TABLE slider decides which depth new points land at.

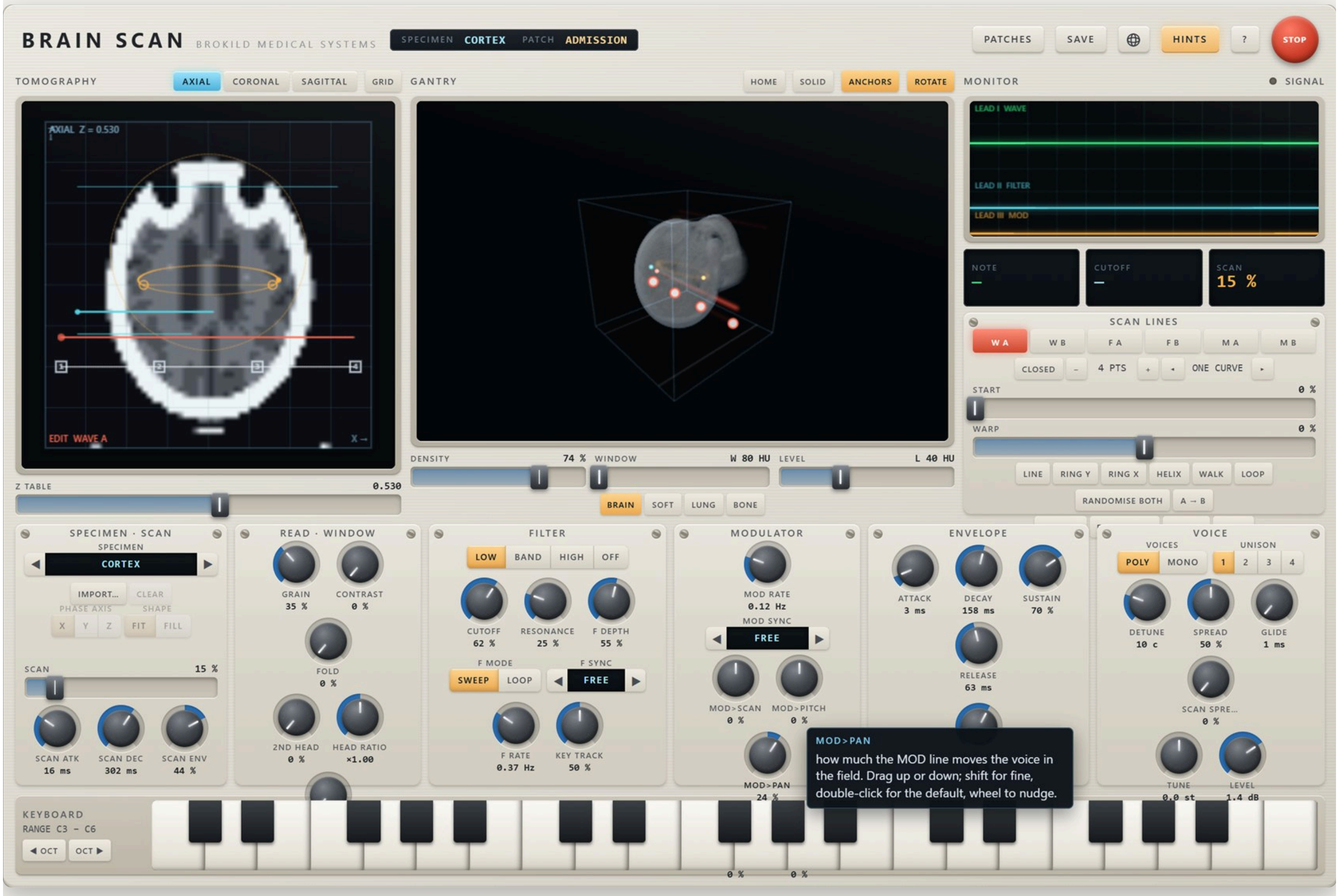
Give SCAN somewhere to go. RANDOMISE BOTH draws fresh walks for both anchors of the selected scanner. **A → B** copies A onto B so you can nudge B away from a shape you already like.

The four regions

Left, the **tomography**: one plane through the specimen, and the only place lines are edited. Middle, the **gantry**: the specimen in three dimensions with the lines glowing inside it — for looking, not for editing. Right, the **monitor** and the **scan lines** bench. Below, the **desk**: specimen and scan, filter, modulator, envelope, voice.

> **Every control says what it is.** Hover anything — a knob, a button, a tab, the slice itself — and a note appears beside it saying what it does and how it is used. The **HINTS** button in the header turns them off; so does the `?` key.

If it goes wrong. The red mushroom stops every voice. Double-click any knob for its default. **PATCHES** holds sixteen factory studies and everything in your patch folder.



The console. Tomography left, gantry centre, monitor and scan lines right, the desk below.

03 The fifteen specimens

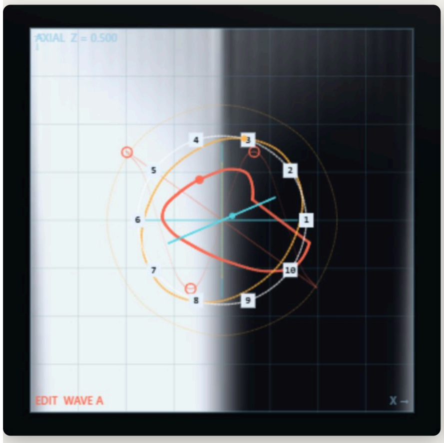
The `x` axis is the phase axis. A straight line along `x` reads the specimen's natural waveform at whatever `(y, z)` it sits at, so an ordinary wavetable is a special case: a straight line, moved sideways. The other two axes are the timbre axes and each specimen says what they mean.

SPECIMEN	A STRAIGHT LINE READS	Y	Z
SINUS	a sine	amplitude	a touch of second harmonic
SPINE	a harmonic stack	brightness	parity: saw to square
PULSE	a pulse	width, 5–95 %	edge softness
MARROW	a ramp	curvature	folds toward a triangle
NERVE	a phase-modulated sine	index	modulator ratio
RETINA	two decaying formants	first formant	second formant
THORAX	a chest CT	ribs, sternum, spine, two lungs with their vessels, the heart and aorta	
SKULL	the skull alone	vault, orbits, sinuses, cheekbones, jaw and teeth — no soft tissue	
CORTEX	a head CT	the same skull with brain, ventricles, eyes and scalp	
SUTURE	a random staircase	steps per cycle, 4–32	walks eight patterns
ENAMEL	a comb of spikes	their width	how many per cycle, 1–8
TENDON	four partials	harmonic to a bell's ratios	brightness
VERTEBRA	three lumbar levels	bodies, discs, the canal and its roots, processes, the muscles around	
FEMUR	a thigh	the femoral shaft flaring into its ends, marrow, muscle compartments, vessels	
JAW	the mandible	both rows of teeth with enamel, dentine and pulp; the tongue and the palate	

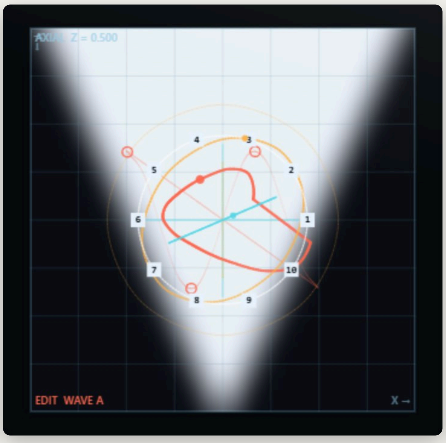
Musically normal by construction. SPINE at a given `(y, z)` is exactly a harmonic series with a known rolloff and a known parity; PULSE at a given `y` is exactly a pulse of a known width. The bench checks both against their formulas — the duty cycle lands within 0.001 of what the formula says. SPINE is the instrument's bread.

The bodies are anatomy, in Hounsfield units. Air is −1000, fat about −90, brain and muscle around 40, trabecular bone a few hundred, the cortical tables above a thousand and enamel at the top of the scale; the cube's 0 to 1 spans −1000 to 2000 HU. So WINDOW and LEVEL read in HU on a body, the four presets on the gantry are a radiographer's, and a bone is a bone: a line through CORTEX reads scalp, skull, cerebrospinal fluid, folded cortex, a ventricle — a waveform with hard edges, a smooth stretch, a burst of detail and a plateau, and moving the line an inch changes which of those it meets and in what order. SKULL is the same head with every soft tissue removed, the way a bone reconstruction shows it. None of them is designed to be tidy. They are designed to be somewhere.

The gritty three are made of edges. SUTURE is a random staircase — `y` the number of steps per cycle, `z` a walk through eight patterns — and a step is a hard edge held for a while: every harmonic, then nothing until the next. ENAMEL is a comb of spikes, `z` how many per cycle and `y` their width; one spike a texel wide is the buzziest thing in the instrument. TENDON is four partials whose ratios slide from harmonic to a bell's. Cut to one cycle their content is harmonic, as every single-cycle read must be; what they carry is the strike at the wrap. Measured above the eighth harmonic, against the fundamental: SPINE's middle −25 dB, SUTURE +4, ENAMEL +10.

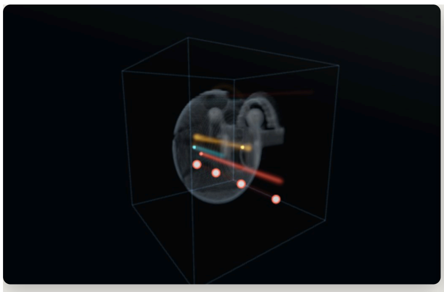


SPINE — a harmonic stack, sliced



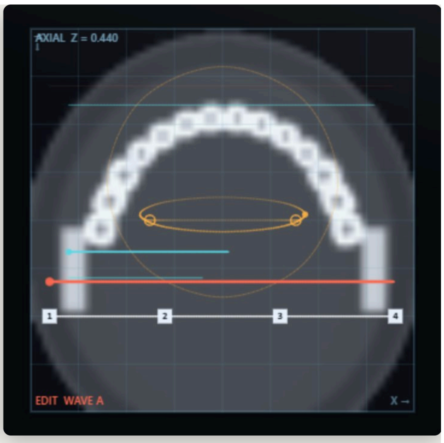
PULSE — width across y

CORTEX — an axial slice, brain window



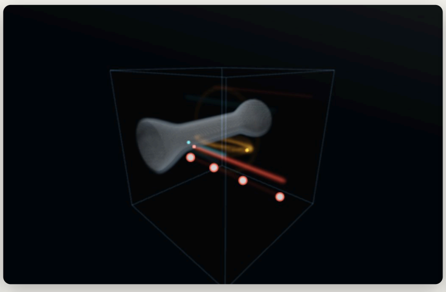
SKULL in the gantry, bone window

THORAX — axial, lung window



JAW — axial through the lower teeth

VERTEBRA — axial through a body



FEMUR in the gantry

SKULL — coronal through the orbits

How the tissue is read

The field is stored as texels and a read lands between them, so a kernel decides what a point between texels is worth. **GRAIN** is that kernel. At 0 it is a smoothing B-spline, which rounds every corner and takes the top of the texel band down by up to 16 dB — the soft read the first build shipped with. At 1 it is an interpolating Catmull-Rom, which passes every texel as it is, kinks included. It is not a tone control: on a bright SPINE, harmonic 20 moves by a decibel between the two and harmonic 40 by five.

CONTRAST is a CT window applied to the sound. Narrowing it scales the read up — to sixteen times — and saturates it against the window's edges: a sine leans toward a square, and there is more for the filter to bite on. **FOLD** is what the window does past its edges: at 0 it clips, at 1 it folds the wave back in, which adds harmonics the tissue never had. Both are anti-derivative anti-aliased; the hardest window at C5 on the brightest field aliases at −51 dB and a full fold at −65. At CONTRAST 0 and FOLD 0 the window is not there at all, and the plain read stays the plain read.

And the volume itself doubled: 128 texels across where the first build had 64. A straight line across the cube carried at most 32 harmonics; now 64. Every specimen was rebuilt for the resolution — PULSE's edge is half a texel wide at the bottom of `z`, SPINE reaches 48 harmonics and a brightness past a saw. The panel still renders 64^3, the same bytes averaged; the engine reads all 128^3.

READ · WINDOW — GRAIN, CONTRAST and FOLD

A volume of your own

IMPORT, under the specimen window, reads a real volume off disk and uses it instead of the fifteen. It takes a **NIfTI** file (`.nii` or `.nii.gz`), a folder of **DICOM** slices, or a folder of images. Any volumetric data will do — a CT or MR of a head, a micro-CT of a fossil, a confocal stack, an industrial scan of a casting.

It does not join the fifteen: it *overrides* them. The window says IMPORTED while it is in use; **CLEAR** gives the dial back, and so does simply turning the dial — a dial that moved without changing anything read as broken, so now it wins. Adding an entry to the dial re-points every patch anyone has already saved, which is why the specimens that arrived in 260904.3 and 260905.1 came with a one-time remap of older patches and projects, and why an import still does not join the dial. An import has Hounsfield units of its own (its cube spans the window it was read with), so the presets light on it too.

What it does to the data, and why

It resamples in millimetres, not in voxels. A clinical CT is typically 0.5 mm across a slice and 1–5 mm between them. Resampling by index would squash the body along the slice axis by exactly that ratio. The spacing is read from the file, the physical extents are fitted into the cube, and a sphere stays a sphere.

It averages when it shrinks. 512 × 512 × 300 into 64^3 is about 8 × 8 × 5 source voxels for every destination voxel. Point-sampling that is aliasing, and aliasing introduced here is in the specimen for good — no later filtering can reach it. Each axis is resampled with a filter as wide as that axis's own ratio.

It picks the window from the data. A single surgical clip at 3000 HU would otherwise push every brain voxel into the bottom two per cent of the range. The window is a percentile of the data rather than its extremes, and the values it chose are printed under the strip in the file's own units — Hounsfield units, for a CT.

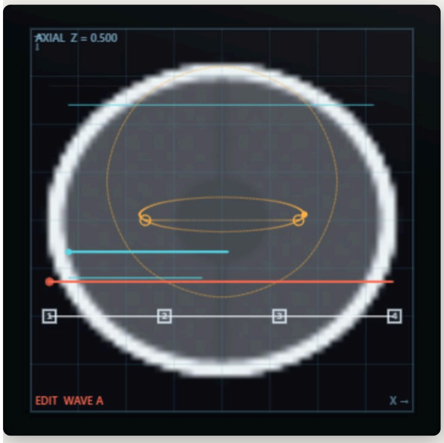
The two controls

PHASE AXIS chooses which axis of the body is read as one cycle. A head read left-to-right, front-to-back and foot-to-head are three quite different instruments. **SHAPE** chooses FIT, which keeps the body's real proportions and pads the rest of the cube with air — silence at those phases — or FILL, which stretches it to the cube. Both re-read the file, because they change how it is read rather than how it is shown.

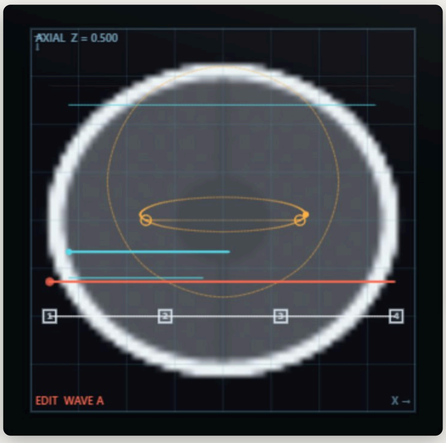
> **The window controls are now literal.** With a CT loaded, WINDOW and LEVEL are a real radiographer's window width and level, in Hounsfield units: air is −1000, cerebrospinal fluid about 0 to 15, grey matter 35 to 45, bone 300 and up. A brain window is about 80 wide centred on 40, and it is the only way to see cortical detail — at a window wide enough to hold the skull, the tissue is all one grey.

An imported volume is *saved with the patch and with the project*, compressed, so a patch is still the whole machine when the folder it came from has been tidied away.

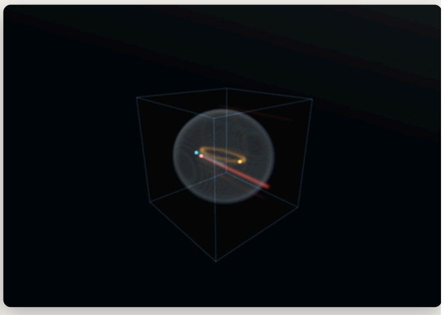
IMPORTED in the window, and what the file turned out to be underneath.



An imported scan, at the window the data chose



The same slice, narrowed to a brain window



And in the gantry, with the lines inside it

04 The three scanners, and the one SCAN

SCANNER	READS	RATE
WAVE	the audio waveform	once per cycle of the note
FILTER	the cutoff of the filter	once per note (SWEEP), or looping (LOOP)
MOD	a modulator: scan, pitch, pan	looping, free or on the host clock

Each has two anchors, A and B. **SCAN** is where between them every scanner is read, and it moves all three together. It has three sources that add up:

- the **SCAN** slider, the base position;
- the **scan envelope** — SCAN ATK, SCAN DEC and SCAN ENV, which is bipolar, so the envelope can travel toward B or back toward A;
- **MOD>SCAN**, the modulator's own contribution.

Because the blend is of geometry, a scan envelope is not a filter sweep and not a crossfade. It is the reading head travelling through the specimen while the note sounds, and what it finds on the way is whatever is there.

Why the anchors matter more than the knobs. Two anchors that lie on top of each other give a SCAN that does nothing. Two anchors in different rooms of the specimen give a SCAN that is the most powerful control on the panel. RANDOMISE BOTH exists for this reason, and it is the fastest way to find out what a specimen contains.

SCAN 0 — read at anchor A

SCAN 50 % — a line neither anchor draws

SCAN 100 % — read at anchor B

The bright line is what is being read. The two faint ones are the anchors; the numbered squares are the control points of the line being edited.

05 Drawing a line

A curve in a cube cannot be drawn with a mouse — but a curve on a plane can, and a plane can be driven through the cube. That is what the tomograph is for, and it is the only place lines are edited.

The plane

AXIAL, **CORONAL** and **SAGITTAL** cut the volume on XY, XZ and YZ. The **TABLE** slider under the slice drives the plane through the specimen; so does the wheel over the slice. The heading in the corner always says which plane and which depth.

The gestures

- **Drag a numbered point** to move it in this plane. *Its depth is kept* — you are moving it on the plane, not to the plane.
- **Click bare tissue** to add a point at the plane's current depth, inserted into the segment it is nearest to.
- **Alt-click a point** to remove it. Two is the minimum.
- **Wheel** moves the table. Hold shift for one texel at a time.

Points off the plane are drawn hollow and fade with distance, so you can always see where the rest of the line has got to. Every line's crossing of the current plane is ringed in that scanner's colour — red for WAVE, cyan for FILTER, amber for MOD.

Split, flatten, undo — and the gantry

SPLIT (the ◂ ▸ pair on the line bench) makes a line two segments instead of one curve: the first half of the cycle reads the points before the split, the second half the points after it. One cycle, two edges, both band-limited — a family of timbre a single curve cannot make. It needs four points or more, and the slice draws the two segments apart.

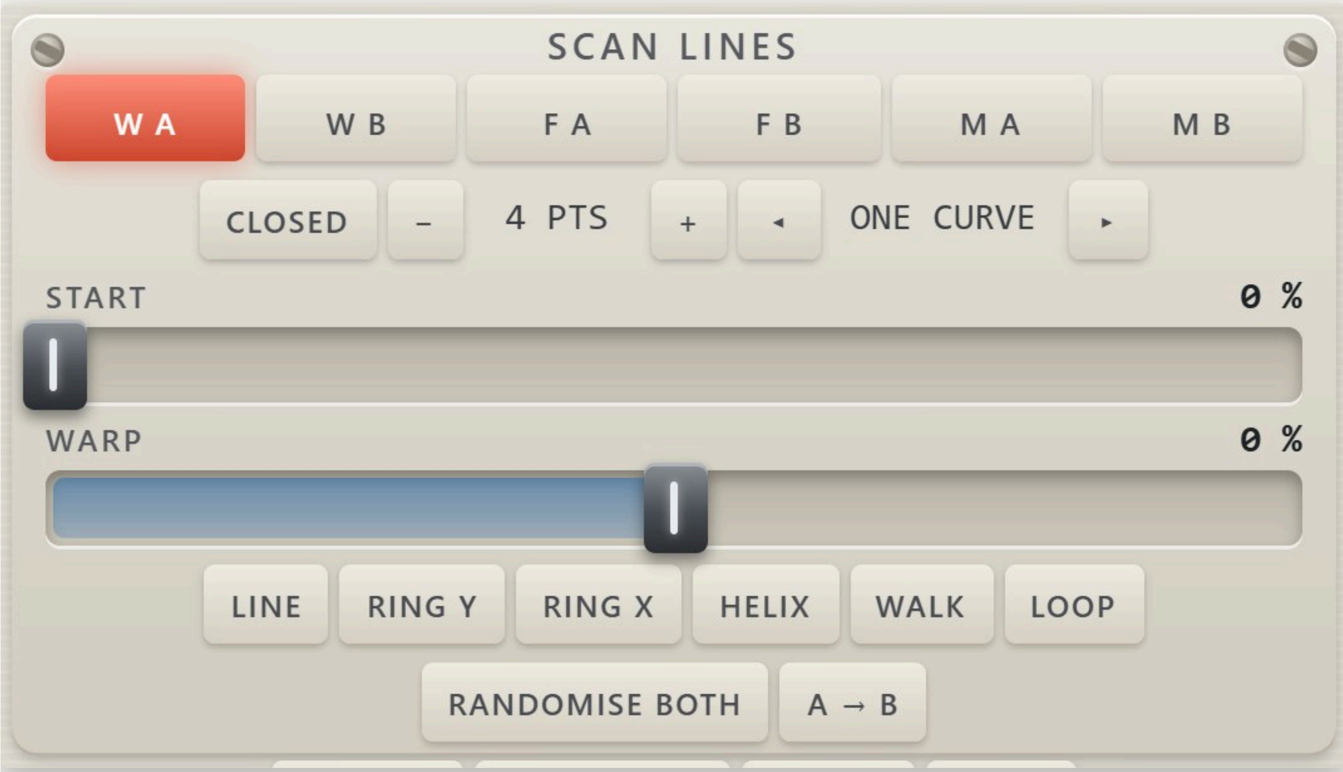
FLATTEN starts the selected scanner over — two straight lines along x through the middle of the body, A a little below B so SCAN still has somewhere to go — and **FLATTEN ALL** does all six: the simplest state the instrument has, to build up from again. **REVERT** puts back the six lines the current patch, study or project was loaded with. **UNDO** takes back the last change to the lines, one step, from any gesture on the slice, the gantry or the bench.

And the gantry is no longer only for looking: the selected line's control points are drawn on it as white dots. Drag one to move it across the view; hold **shift** to push it along the line of sight — the one direction the slice cannot reach.

The bench

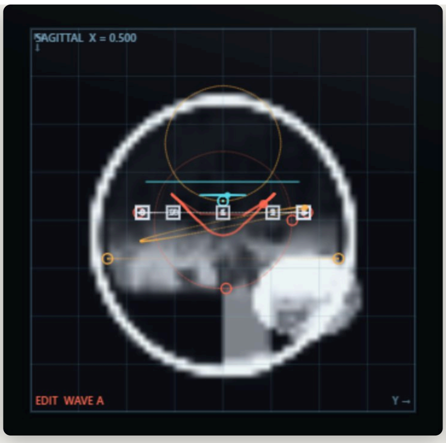
The **SCAN LINES** panel on the right chooses which of the six lines you are editing and carries its own properties.

CLOSED	loop or segment — no edge, or one edge per cycle
– / +	take a point off the end, or add one; two to sixteen
START	where a closed loop's cycle begins: moves the phase, not the shape
WARP	phase distortion — the reader runs faster over one half of the line
LINE … LOOP	lay a shape through the line's present centre. WALK and LOOP are new every time you press them
RANDOMISE BOTH	fresh walks for both anchors of this scanner
A → B	copy this scanner's A anchor onto its B anchor

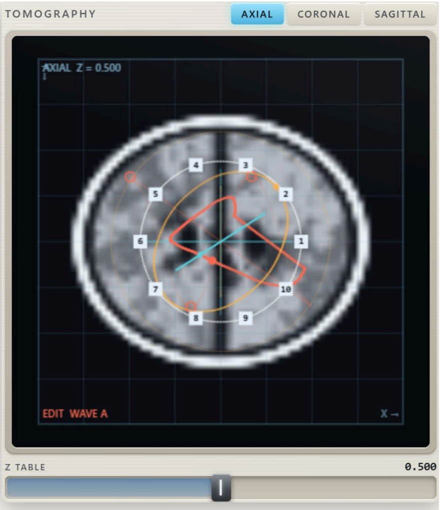


The scan-line bench. Six lines, and the tools for the one you are editing.

CORONAL — the same lines, cut on XZ



SAGITTAL — cut on YZ



The tomograph and its table

06 The gantry

The middle screen is the specimen in three dimensions, ray-marched, with the scan lines glowing inside it. It is for looking: nothing is edited here. Drag to turn the specimen, wheel to come closer, **HOME** to put it back, **ROTATE** to let it turn on its own.

The bright line is what is being read at the present SCAN; the two dimmer ones are its anchors, and **ANCHORS** hides them. The small travelling dots are the reading heads — the audio-rate one shown slowed down, because nothing about 300 revolutions a second is watchable.

DENSITY, WINDOW and LEVEL

These three are the transfer function, and they are the scanner's own vocabulary.

- **DENSITY** is opacity. Low, and the specimen is a ghost the lines blaze through; high, and the tissue closes over them.
- **WINDOW** is the window's *width* — how much of the density range is shown at all.
- **LEVEL** is the window's *centre* — which density it is looking at.

WINDOW and LEVEL also drive the slice, exactly as a window and level pair drives a real radiograph. They are set for you when a specimen loads, from that specimen's own histogram, and then stay where you put them until the next one.

The presets, in Hounsfield units

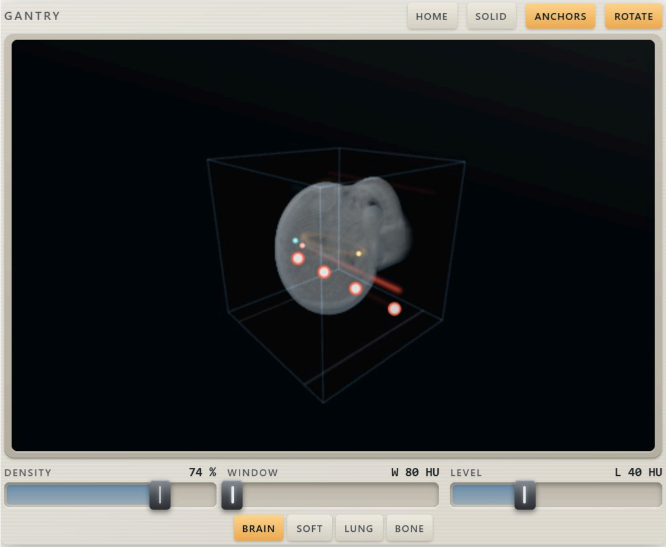
Under the sliders sit **BRAIN** (W80 L40), **SOFT** (W400 L40), **LUNG** (W1500 L−600) and **BONE** (W2000 L500) — a radiographer's windows. They light on the bodies and on an import, both of which carry real CT numbers, and each body opens on its own: the head on BRAIN, the chest on LUNG, the bones on BONE. A preset also raises DENSITY, because bone is a thin shell and the gantry needs opacity to show it; and on a body the gantry draws nothing below a third of the window, so a bone window shows bone and not a fog of soft tissue. The phantoms have no HU: the buttons dim, and the sliders work as they always did.

SOLID and SURFACE

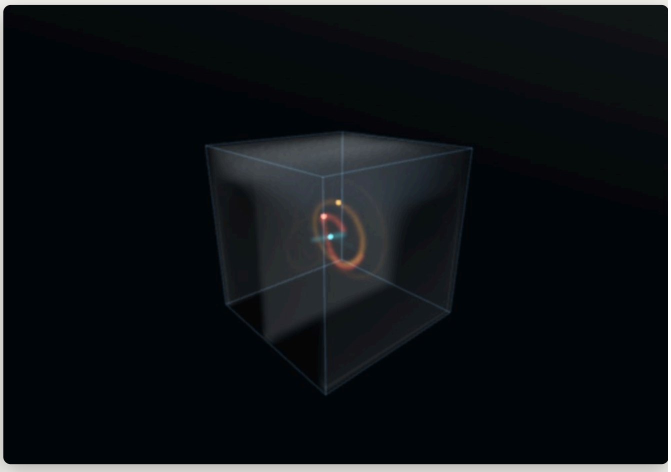
A specimen with air around it and a specimen whose field fills the cube want opposite treatments, so the treatment is a button.

- **SOLID** shows everything denser than the window's floor. That is how a scanned object should read: the brain, the skull.
- **SURFACE** is opaque only in the middle of the window, so a field that fills the cube becomes the single surface where it crosses the level. Sweep LEVEL in this mode and you travel out through the specimen one shell at a time.

It is chosen for you when a specimen loads — SOLID if more than a third of the cube is empty — and you can override it.



CORTEX in SOLID, the lines glowing through



SPINE in SURFACE — one level of the field

07 The filter, the modulator, the voice

FILTER

A state-variable filter with **LOW**, **BAND**, **HIGH** and **OFF**. CUTOFF is where it sits when the FILTER line has no say; **F DEPTH** is how much of the say the line has. At F DEPTH 0 the filter is an ordinary filter. Turn it up and the cutoff becomes a reading of the specimen along the FILTER line — which means a filter sweep with the shape of whatever the line passes through, not a curve from a table.

F MODE chooses **SWEEP**, one traversal per note, or **LOOP**, going round for as long as the note lasts. **F RATE** is the sweep time or the loop rate, and **F SYNC** locks the loop to the host clock. **KEY TRACK** is how much the cutoff follows the note.

MODULATOR

A third line, read at **MOD RATE** (or on the host clock through **MOD SYNC**), whose reading is sent three places: **MOD>SCAN**, **MOD>PITCH** and **MOD>PAN**. All three are bipolar. A modulator whose line loops through a rough part of the specimen is a very different animal from one that loops through a smooth part; the shape of the modulation is the shape of the tissue.

The second head, and the MOD line on the read

2ND HEAD puts a second reader on the same blended line and mixes it in. At **HEAD RATIO** ×1.00 with **HEAD PHASE** offset it is a fixed-interval double without a second voice — at 50 % it cancels the odd harmonics and leaves a hollow. Off the whole numbers it is something a single-cycle read can never be on its own: a partial that is not a harmonic. At ×1.50 the bench measures a partial at 1.5 f_0 within 0.3 dB of the fundamental and no second harmonic at all. Whole ratios give harmonics; anything else beats.

MOD>WINDOW and **MOD>GRAIN** let the MOD line drive the read itself: the tissue under the MOD line narrows the CONTRAST window or sharpens GRAIN as it travels, so the body decides how hard it is read. Centred, they do nothing — the bench checks the knob values pass through untouched. **SCAN SPREAD**, in VOICE, spreads unison readers through the SCAN as well as in pitch: each reads the body a little further along than the last, and a chord of readers widens without detuning.

ENVELOPE and VOICE

ATTACK, DECAY, SUSTAIN and RELEASE with VELOCITY sensitivity, and the usual voice controls: **POLY** (eight voices) or **MONO** with legato glide, **UNISON** of one to four readers per note, **DETUNE** and **SPREAD** for them, **GLIDE**, **TUNE** and **LEVEL**.

> **Unison is literal here.** Four readers walk the same line at slightly different rates. They are reading the same tissue, so they stay related to one another however strange the line is.

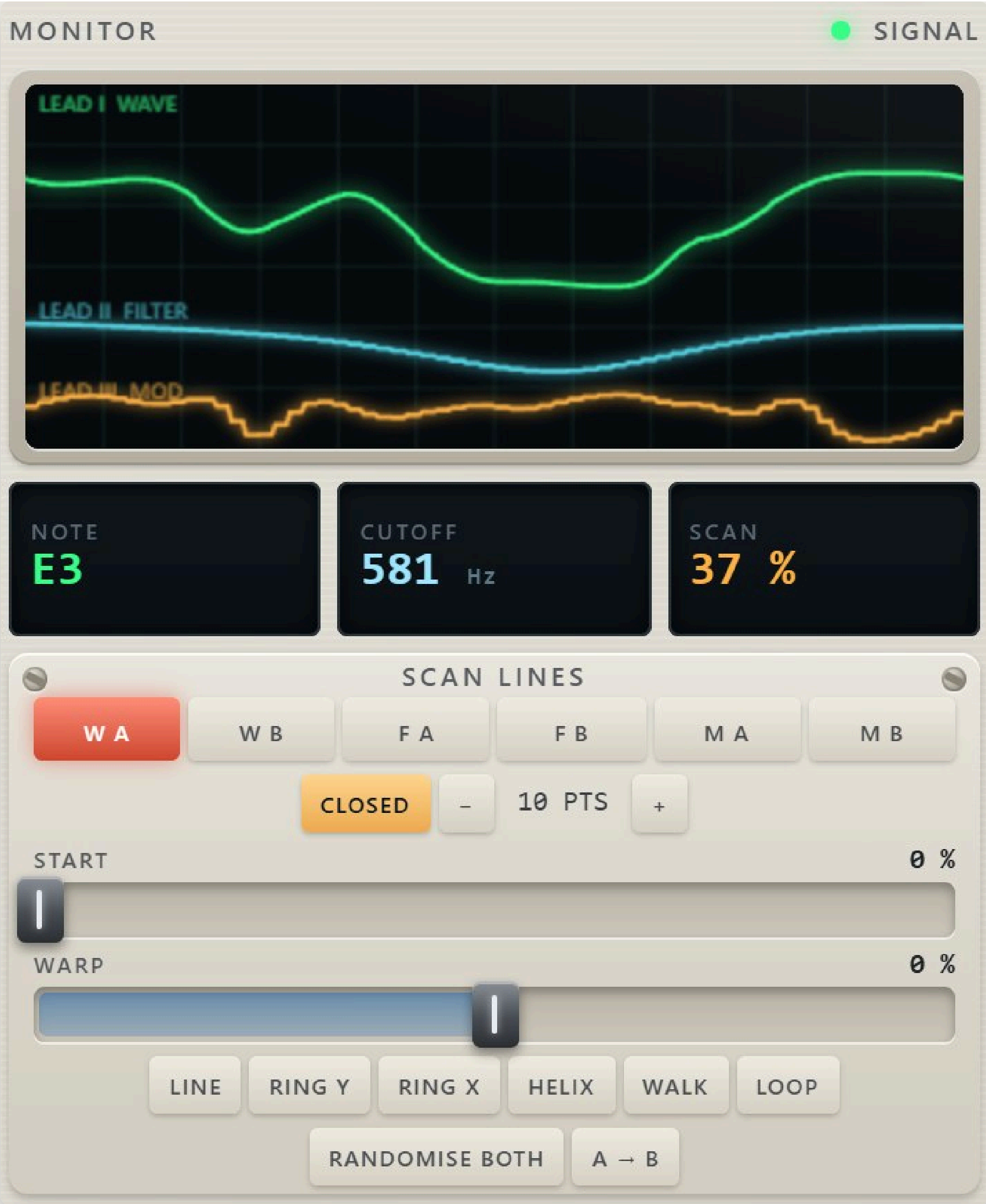
The monitor

Three leads: **LEAD I** is the cycle the WAVE scanner is reading, **LEAD II** the FILTER line, **LEAD III** the MOD line. Under them, the note being played, the cutoff in hertz, and where SCAN has actually got to — which is not the same as where the slider is, once the envelope and the modulator have had their say.

FILTER

MODULATOR

VOICE



The monitor, the vitals, and the scan-line bench.

08 Patches, and the world rack

PATCHES in the header holds sixteen factory studies and everything in your patch folder. A patch is every control, both anchors of all three scanners, the specimen and the rack — the whole machine. **SAVE** writes one.

Patches live in `Documents\Brokild patches\Brain Scan`, the same house folder every Brokild plug-in uses. You can point the plug-in somewhere else from the PATCHES menu, and it will remember. A patch is plain JSON; it can be sent to somebody.

ADMISSION	where a fresh instance opens
SINUS RHYTHM	a sine that will not sit still
CORTEX	the brain, played as an instrument
PULSE OX	width where width is not a knob
MARROW	the ramp, folded
NERVE	phase modulation read as a landscape
RETINAL	two formants and a scan between them
VENTILATOR	slow, breathing, on the host clock
ARRHYTHMIA	a line that should not work
STAPLES	the staircase, read texel by texel
ENAMEL	spikes through a narrow window
TENDON	bell ratios, folded
SKULL	through the orbits, then through the teeth
VERTEBRA	a second head at ×1.50 — a beating partial
FEMUR	a split line: bone, then muscle, two edges a cycle
JAW	the MOD line narrowing the window as it goes

Brokild World FX

The globe in the header opens the shared rack every Brokild instrument carries: twelve effect modules and four character modules, dragged into whatever order you like, and five **MACROS** that appear as host parameters so the rack can be automated. A fresh rack is empty and changes nothing.

> **Five macros, and they map rather than add.** A macro sweeps its destination across the destination's whole range, so a macro at 100 % is that control at its top and 0 % is its bottom. MACRO 5 arrives assigned to the rack's dry/wet, which is why a fresh instance sounds exactly as it would with no rack at all.

Brokild World FX, over the console.

09 What is measured

Nothing in this manual is a claim somebody thought sounded right. The engine has an offline bench that renders real audio and asserts on it — 118 checks — and the panel has a second one that loads the real page and drives it through the plug-in's own message vocabulary — 43 checks. Both are in the source tree.

WHAT	MEASURED
SINUS is a sine	total harmonic distortion −120.9 dB
SPINE obeys its formula	centroid rises with y; at z = 1 the even harmonics sit 47.4 dB under the odd
PULSE obeys its width	duty 0.229 / 0.500 / 0.771 against a formula saying 0.230 / 0.500 / 0.770
Aliasing	the roughest specimen (THORAX, grain and all) at C6: non-harmonic energy −70.1 dB, and +0.6 dB with the mip pyramid disabled
The edge of an open line	−53.4 dB with the band-limiting step, −20.0 dB without: it is worth 33 dB
Offset	−114.0 dB of DC after a second
The claim	the fifth harmonic at SCAN 50 %: −13.6 dB, the formula −14.0; the crossfade of the same two anchors −61.2
The filter follows its line	worst error 0.000 octaves against the field read along the blended line
Cost	16.7 % of one core for eight voices of four-way unison — 32 readers, the maximum; 18.5 % with GRAIN 1, the window and the fold all on; 39 % with a second head on every reader
The second head	at ×1.00 and PHASE 50 % the odd harmonics fall from +3.9 dB over the even to −106 dB under; at ×1.50 a partial at 1.5 f_0 sits 0.3 dB from the fundamental with the 2nd harmonic at −111 dB; at ×4 on the brightest field at C5 the floor is −56 dB
A split line	reads its first segment to the end and jumps to the second at exactly half a cycle; at C5 the two edges band-limited give −64.3 dB of non-harmonic energy, −31.8 without — the second polyBLEP is worth 33 dB
The MOD line on the window	centred, SINUS reads −120.6 dB of THD; at −100 % with the MOD line parked in the sine's trough, −8.5 dB
The bodies	tissue fractions per body (air where there is air, a few per cent of bone, enamel where there are teeth); five rays from the centre of the head all cross bone and the SKULL's cranium is empty while CORTEX's holds white matter; the chest has two lungs; the vertebra has a canal of CSF between two pedicles of bone
GRAIN	SPINE at y = 1, A2: harmonic 40 rises 4.6 dB from GRAIN 0 to 1 while harmonic 20 moves 1.3 dB — a kernel, not a tilt
CONTRAST	SINUS THD −120.6 / −18.3 / −10.3 / −7.9 / −6.9 dB at CONTRAST 0 / ¼ / ½ / ¾ / 1: nothing at zero, monotonic to a square's order
The window's aliasing	brightest SPINE, GRAIN 1, C5: −58.3 dB plain, −51.2 dB with CONTRAST 1, −64.9 dB with a full FOLD at gain 4
The gritty three	energy above the 8th harmonic against the fundamental: SPINE's middle −24.6 dB, SUTURE +4.4, ENAMEL +10.1, TENDON −2.6
The build	a 128^3 specimen builds in 21 ms on twenty threads; the panel's copy is published from a worker, never from the timer
The import	a volume that is 80 × 80 × 20 voxels but physically cubic fills the cube, and a sphere in it still measures the same on all three axes — resampling by index would read about 12 voxels on the coarse one
Decimation	a one-voxel checkerboard, decimated 4:1, flattens to standard deviation 0.00005; point sampling would leave 0.5
No shift	a ramp resamples to a ramp within 0.006 — the half-voxel error that breaks a resampler without looking broken
Outliers	with one voxel at 30000 among values of 0 to 100, the window comes out 0 to 103
Readers	NIfTI and DICOM round-tripped from bytes built in memory: dimensions, spacing, rescale to Hounsfield units, and slices ordered by their position rather than the order they arrived. Compressed DICOM and NIfTI-2 are refused by name.

One number is worth reading twice. The mip pyramid — reading the volume at a coarseness matched to the pitch and the length of the path — is worth 44 dB of aliasing at C6. A volume read naively at high pitch is not a slightly rougher instrument; it is a broken one.

The panel probe checks, among other things, that a straight line of four collinear control points reads exactly straight — that the panel evaluates a curve the same way the engine does, to five decimal places. What you see drawn is what you hear.

10 Every control

SPECIMEN · SCAN

SPECIMEN	which volume the lines read. Not automatable: it rebuilds two million texels on a worker thread, and a lane fighting that is unusable
SCAN	where between A and B every scanner is read
SCAN ATK	how fast the scan envelope rises
SCAN DEC	how fast it falls back
SCAN ENV	how far the envelope moves the scan, either way ±100 %

READ · WINDOW

GRAIN	the read kernel: 0 smooths across texels, 1 passes every texel as it is
CONTRAST	the CT window on the sound: narrower saturates the read toward a square, up to sixteen times the gain
FOLD	past the window's edges: 0 clips flat, 1 folds the wave back in
2ND HEAD	a second reader on the same line, mixed in
HEAD RATIO	its speed against the note, ×0.25 to ×4 with exactly ×1.00 in the middle
HEAD PHASE	where along the cycle it starts — the interval, at ×1.00

THE READ, MODULATED

MOD>WINDOW	how much the MOD line narrows the window (in MODULATOR)
MOD>GRAIN	how much the MOD line sharpens the read (in MODULATOR)
SCAN SPREAD	unison readers spread through the scan, each a little off the others (in VOICE)

FILTER

FILTER	LOW, BAND, HIGH, OFF
CUTOFF	the cutoff when the FILTER line has no say
RESONANCE	the filter's resonance
F DEPTH	how much the FILTER line decides the cutoff
F MODE	SWEEP once per note, or LOOP
F RATE	the sweep time, or the loop rate, 0.05–40 Hz
F SYNC	FREE, or 4/1 down to 1/16 and the triplets
KEY TRACK	how much the cutoff follows the note

MODULATOR

MOD RATE	how fast the MOD line is read, 0.02–30 Hz
MOD SYNC	lock it to the host clock
MOD>SCAN	how much it moves the scan, ±100 %
MOD>PITCH	how much it bends the pitch
MOD>PAN	how much it moves the voice in the field

ENVELOPE

ATTACK · DECAY · SUSTAIN · RELEASE	1 ms to 10 s on the timed three
VELOCITY	how much velocity decides the level

VOICE

VOICES	POLY (eight) or MONO with legato glide
UNISON	one to four readers per note
DETUNE	how far they are spread in pitch, 0–40 cents
SPREAD	how wide they sit in the stereo field
GLIDE	portamento between notes
TUNE	master tuning, ±12 semitones
LEVEL	output level

IMPORT

IMPORT...	read a NIfTI file, a DICOM folder or a folder of images
CLEAR	throw it away and give the SPECIMEN dial back
PHASE AXIS	which axis of the body is one cycle of the waveform
SHAPE	FIT keeps the body's proportions; FILL stretches it to the cube

The screens

AXIAL / CORONAL / SAGITTAL	which plane the tomograph cuts
GRID	show the 128 texels the engine reads across the plane
BRAIN / SOFT / LUNG / BONE	a radiographer's windows, on a body or an import
SPLIT ◂ ▸	make the line two segments; move the split; back to one curve
FLATTEN / FLATTEN ALL	the scanner, or every line, back to straight reads through the middle
REVERT	the six lines as they were loaded
UNDO	the last change to the lines, one step
TABLE	the depth of that plane
DENSITY	the gantry's opacity
WINDOW / LEVEL	the width and centre of the display window, both screens
SOLID / SURFACE	how the window is read in the gantry
HOME / ROTATE / ANCHORS	the camera, its spin, and whether the anchors show

11 Installing, and the small print

Windows, 64-bit. The download holds a VST3 and a standalone. Put `Brain Scan.vst3` in `C:\Program Files\Common Files\VST3\` — a subfolder is fine, hosts scan recursively — and rescan in your host. The standalone needs nothing but itself.

The panel is a web view. It needs the Microsoft Edge WebView2 runtime, which is present on every up-to-date Windows 10 and 11. If the panel comes up blank, that runtime is what to install.

It is not signed. Windows may say so on first run, and Smart App Control may refuse an unsigned binary outright; that is a Windows setting, not a fault in the file.

Host notes

- Thirty-nine parameters are automatable, plus five rack macros. SPECIMEN is deliberately not: it rebuilds the volume.
- The scan lines are saved with the project and with a patch, but they are not host parameters. A curve of sixteen points in three dimensions is not a fader.
- F SYNC and MOD SYNC take the host's tempo. With no host clock they fall back to their own rate.
- MIDI: notes, velocity, pitch bend, sustain, and the modulation wheel.

If something is wrong

- **No sound.** Check the specimen actually has tissue where the WAVE line is — the slice will show you. A line entirely in air reads silence, correctly.
- **A patch sounds different after reloading.** It should not; if it does, that is a bug worth reporting.
- **Everything is buzzing.** RESONANCE with F DEPTH high can find a place in the tissue that sweeps the cutoff faster than it can be believed. That is the instrument working. LEVEL and the limiter in the rack are both downstream.

> **Brokild plug-ins are free.** No licence, no key, no phoning home. They are made for the pleasure of making them.